

Project contribution report

Project: NBP Analyzer

Team: [ZPI2025_IO3_BunniHopPon](#)

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Core Logic & CI Setup

Histogram Distribution Module GUI:

Integrated the histogram distribution module into the React SPA front-end, connecting the change distribution logic to the histogram chart component.

Statistical Analysis Module GUI:

Integrated the statistical analysis module (min, max, avg, median, mode, standard deviation, coefficient of variation) into the React SPA. Implemented the statistical badges panel displayed below the session analysis chart, ensured correct period selection binding.

Activity Diagram:

Created the UML Activity Diagram for the project documentation, covering the main application workflows: data fetch and validation flow, session trend calculation flow, statistical measures calculation flow, and histogram generation flow. The diagram is part of the project documentation stored in the repository and reflects the final implemented behaviour.

Bug Fixes

Missing Validation for Same Currency Selection:

Identified and resolved a bug in the change distribution view where the GUI allowed a user to select the same currency for both inputs of a currency pair (e.g. EUR/EUR). The fix adds front-end validation that dynamically disables the already-selected currency in the second selector, preventing invalid pair submission.

Incorrect Business Logic for Currency Pairs:

Diagnosed and fixed incorrect cross-rate calculation logic in the change distribution module. The system was computing pair changes by directly subtracting PLN-based rates instead of deriving the correct cross-rate (base/quote). Corrected the formula to dynamically calculate cross-rates from

PLN-denominated data retrieved from the NBP API, consistent with the system constraint that all base rates are in PLN.